Knowledge Architecture of Ensha

Version 0.2

Part One — Philosophy of Knowledge Architecture

Introduction

Ensha ("Pencil of Thought") is not merely a system for producing educational content. Nor is it a system for analysis or student knowledge assessment. Ensha is an educational learning ecosystem that works over time to provide more favorable conditions for the emergence of "thinking movements" in the classroom. For this reason, what is produced, stored, and used in Ensha is not just data; rather, it comprises different layers of knowledge that take shape through the interaction between the teacher, the classroom, and artificial intelligence.

The Objective of Knowledge Architecture

The goal of this architecture is to answer this question: How is knowledge in Ensha produced, how does it change, and how is it once again placed at the service of education? This architecture explains the flow of knowledge throughout the learning cycle.

Principle One — Knowledge Is the Product of Collaboration

In Ensha, no single component produces knowledge on its own. Knowledge is the result of collaboration among the teacher, the class, students' writings, the discovery engine, the recommendation engine, and class memory. Each component completes a part of the cycle.

Principle Two — Knowledge Is Always Temporary

Ensha never claims to have "understood the class." What is recorded in class memory is always based on evidence, dependent on time, and subject to

revision. With the arrival of new evidence, any part of this knowledge may be corrected, strengthened, or faded.

Principle Three — The Ultimate Goal Is Not to Know the Class

Knowing the class is not the goal of Ensha. Knowing the class is only a tool for better designing future sessions, selecting more appropriate methods, and providing more opportunity for the emergence of thinking movements. Therefore, the value of knowledge in Ensha lies not in "how accurately it describes the class," but rather in "how much it helps improve education."

Principle Four — Knowledge Serves the Learning Cycle

All components of the knowledge architecture have only one function: helping the learning cycle become more complete. Knowledge that has no effect on the quality of educational design has no place in the Ensha architecture.

The Five Layers of Knowledge Architecture

Knowledge in Ensha takes shape in five layers:

- **Thinking Framework** (چارچوب حرکت‌های فکری)
- **Atlas** (اطلس روش‌ها)
- **Discovery Knowledge** (دانش اکتشافی)
- **Class Memory** (حافظه کلاس)
- **Recommendation Knowledge** (دانش پیشنهاد)

Each layer has a different role in the learning cycle and only gains meaning in connection with the other layers.

Layer One: Thinking Framework

Definition

The Thinking Framework is the theoretical core of Ensha. This framework defines a set of thinking patterns that Ensha tries to create opportunities for their emergence in classes during the writing process. All other components of the knowledge architecture are directly or indirectly dependent on this framework.

Position in the Architecture

This layer is the highest and most stable layer of the knowledge architecture. Unlike the other layers, it does not learn from classes; it does not change with the execution of sessions; and it is not rewritten based on educational data. This layer can only change as a result of theoretical research or a decision by the project's scientific team.

Roles

- **Common project language**: All components of Ensha communicate with each other using this same language.
- **Reference for interpreting evidence**: All evidence and hypotheses that are produced must be interpretable within the framework of one of the thinking movements.
- **Educational orientation**: The goal of designing activities is to create an opportunity for the emergence of one or several thinking movements, not to produce a better text.

Relationship with the Atlas

The Thinking Framework and the Atlas are two different concepts: thinking movements are educational goals; the Atlas is an educational tool. There is no one-to-one relationship between these two.

Summary

The Thinking Framework is the stable reference point of the entire knowledge architecture of Ensha. All other components of the project carry out their activity within the scope of this framework.

Layer Two: Atlas

Definition

The Atlas is the library of Ensha's educational methods. Each method provides a different path for designing a writing situation. The goal of the Atlas is not to determine the content of a session; rather, it is to offer diverse ways to create opportunities for the emergence of thinking movements.

Position in the Architecture

The Atlas is the second stable layer of the knowledge architecture. Its methods are independent of classes, teachers, and students' writings. Change in the Atlas only happens as a result of the project's scientific development.

The Nature of Methods

The Atlas's methods are not educational goals, evaluation indicators, or cognitive models; they are merely educational design tools. The success of a session is not measured "by executing a method," but rather is evaluated by the opportunities that method created for the emergence of thinking movements.

Relationship with the Recommendation Engine

The recommendation engine has direct interaction only with the Atlas. The output of the recommendation engine is always one of the Atlas's methods; the recommendation engine does not produce new methods, does not change methods, and does not combine them.

Summary

The Atlas is the educational design knowledge of Ensha. The thinking movements determine what should have the opportunity to emerge; the Atlas determines through which paths that opportunity can be provided.

Layer Three: Discovery Knowledge

Definition

Discovery knowledge is the body of knowledge that the discovery engine produces from the evidence present in the class. This knowledge is obtained directly from students' writings, class performance, and teacher feedback. However, it is not yet considered part of class memory.

The Main Task

Contrary to the common perception of "text analysis" tools, the discovery engine does not have the task of precise diagnosis. Its task is to turn every sign — even weak and uncertain ones — into a plausible hypothesis. Final precision is not the product of this layer; it is the product of the teacher's feedback cycle over time (see the "Ensha Document Foundation" and the 14 Fundamental Laws / Principles).

Two Output Levels of the Discovery Engine

In order for the philosophy of action to remain simple and useful, the discovery engine produces two completely separate output levels that must not be mixed:

Level One — Daily Encouragement (primary output)
From among the writings of each week, the discovery engine identifies a few cases (usually three to five per class) that are truly worth being seen, and expresses them in simple language, without the need for precise classification by thinking movement, and shows them to the teacher. This level shapes the teacher's daily experience with Ensha and directly fulfills the goal of Principle Zero (reducing reading burden, increasing encouragement).

Level Two — Background Enrichment (secondary output)
Findings are framed within hypotheses related to the thinking movements defined in the framework and recorded for the gradual enrichment of class memory. This level accumulates in the background; it is not under the pressure of momentary precision, and it is not directly exposed to the teacher's daily view. This separation also reduces the cost and technical complexity of the discovery engine.

Level One is a general and qualitative judgment ("this resembles something interesting," not a precise classification among twelve close categories). For this reason, according to Principle/Law 36, this layer requires only an intermediate class of language models, not the most complex ones available.

The Principle of Free Hypothesis-Making
The discovery engine is not constrained or conservative in producing hypotheses. If there is any evidence — however weak — for the emergence of a thinking movement, the discovery engine must turn it into a low-credibility hypothesis. This is not because it is certain of hypothesis production, but deliberately: it is better to build ten hypotheses of which three turn out to be real, than to miss just one real opportunity by producing only two "confident" hypotheses. Since no hypothesis is punished (Law 13), this freedom in hypothesis production has no cost to the overall health of class memory — an incorrect hypothesis simply never receives teacher confirmation and fades in review.

Position in the Architecture
Discovery knowledge is the first dynamic layer in the knowledge architecture. Unlike the Thinking Framework and the Atlas, which are defined in advance, this layer is produced in the flow of each session: Class → Students' writings → Discovery knowledge.

Components of Discovery Knowledge

a) Evidence
The smallest unit of knowledge. Evidence is extracted directly from class data and is not yet interpreted.

b) Signals
When several similar pieces of evidence come together, a signal takes shape.

c) Hypotheses
A hypothesis is the educational interpretation of signals and is not yet a definitive judgment.

d) Confidence
Each hypothesis has a confidence score that changes only through two mechanisms: collective strengthening (from new evidence or teacher confirmation) and multiplicative fading (from explicit teacher rejection or the passage of time without feedback). In this architecture there is no negative score; a hypothesis is never invalidated or deleted, it only fades.

The precise details of the formula, coefficients, and thresholds are given in the "Hypothesis Credit Confidence Model" document, which is the final reference.

In calculating the effect of teacher feedback, the history of his or her work with that same group of students (not just the class title) carries different weight; details are given in the fourth layer (Class Memory) and the Credit Model in the document.

Relationship with the Teacher
The discovery engine makes its hypotheses observable to the teacher, but the teacher is not obliged to accept them. Teacher alignment is one of the factors that increases or decreases the confidence of hypotheses, not the source of their creation.

The Principle of Epistemic Humility
Discovery knowledge never claims to have understood the student's personality, discovered the class's stable characteristics, or reached a definitive conclusion. All outputs of this layer are evidence-based hypotheses and always remain open to revision.

Summary
Discovery knowledge is the first dynamic layer of the knowledge architecture of Ensha. In this layer, using the evidence present in students' writings, the discovery engine reveals the daily encouragement-worthy moments to the teacher (Level One) and, at the same time, freely and without excessive caution, produces signals and hypotheses for the gradual enrichment of class memory (Level Two). This layer rests on the principle of "abundance of hypotheses, not precision of the moment," and "evidence, not certainty."

Layer Four: Class Memory

Definition

Class memory is the educational knowledge repository of each class. This memory maintains a relatively stable body of information obtained from the multiple cycles of teaching, discovery, and feedback.

The goal of class memory is not to record the past; rather, it is to help design a better future.

Position in the Architecture

Class memory comes after discovery knowledge. Only the part of discovery knowledge that gains sufficient credibility over time enters class memory: Students' writings → Discovery knowledge → Class memory.

First Principle

Class memory is not the students' memory. In Ensha, there is no memory for describing or classifying students; class memory is maintained only about the class itself.

Second Principle

Only useful knowledge is stored. A piece of data enters class memory only when it helps design better future sessions, is based on evidence, and has passed through the cycle of discovery and feedback.

Teacher Feedback Credit Coefficient

The teacher's role in enriching class memory is not constant; rather, it carries different weight based on his or her history of working with that same group of students (not merely the class title or academic base):

- **First year of collaboration**: first three months coefficient 1, rest of the year coefficient 2
- **Second year of collaboration**: first three months coefficient 2, rest of the year coefficient 3
- **Third year and beyond**: all year coefficient 3

These coefficients have an effect in two separate paths: structured feedback (e.g., end-of-class questions) with a milder weight, and the teacher's free notes with a more colorful weight — because free notes have no specific structure and require more interpretation and experience. A teacher who has worked with the same students for three years or more can, on his or her own, build a full hypothesis or add significant weight to an existing one.

Complete details of the formula are given in the "Hypothesis Credit Confidence Model" document.

Components of Class Memory

1. **Stable data**: Information that usually does not change: academic base, number of students, class facilities, implementation constraints.
2. **Dynamic data**: Information related to the class's current situation: conditions of this week, occasions/occasions, the class's overall situation, current educational goals.
3. **Active hypotheses**: The most important part of class memory. Hypotheses that have sufficient evidence and are still credible, and can influence future recommendations.
4. **Educational experiences**: The result of the class's past experiences: which method has been executed several times, which method has been more effective for this class.

What Is Not Stored in Memory

Student grades, student rankings, individuals' personality traits, psychological judgments, individual labels, and permanent analyses about individuals — this information is outside the epistemic boundaries of Ensha.

The Principle of Dynamism

Class memory is not permanent; each part of it may be strengthened, corrected, or faded. Fading, according to Law 13, never means the forced deletion of a hypothesis — the hypothesis remains in memory and, in the event of new evidence or fresh confirmation, can be strengthened again.

Summary

Class memory is the core of Ensha's long-term learning. This dynamic knowledge, based on evidence and open to revision, is enriched with different weights of teacher feedback based on their history of collaboration, and is maintained with only one purpose: to help design better education for that same class in the future.

Layer Five: Recommendation Knowledge

Definition

Recommendation knowledge is the layer of knowledge that transforms the knowledge in class memory into educational recommendations. This layer does not produce new knowledge; rather, it uses the knowledge previously formed in

the teaching and discovery cycle to recommend the most appropriate path for the next session.

First Principle
The recommendation engine does not discover, does not diagnose, does not analyze, and does not produce memory. It only recommends.

This conceptual simplicity directly leads to the computational simplicity of this layer: because the recommendation engine is merely a rule-based and scoring logic based on existing data, according to Law 36 it requires no complex AI models.

Output of the Recommendation Engine
The output of the recommendation engine is only one thing: a recommendation of an Atlas method. No teaching situation is created at this stage.

The Principle of Humility
The recommendation engine never claims that its recommendation is the best or the only correct method; it merely says that, given the knowledge now available about this class, this method is the most appropriate current recommendation.

Summary
Recommendation knowledge is the last layer of the knowledge architecture of Ensha. The quality of these recommendations improves over time, not through changes to the recommendation engine, but through the gradual growth of class memory and the learning cycle.

End of Document

This is version 0.2. The first and second layers are stable and, according to their nature, do not change; the third, fourth, and fifth layers will, along with real-world implementation experience and refinement — in accordance with the principle — be subject to revision.